as oil, wax, fat and dust are removed to produce hydrophilic and clean textile material. It is one of the vital processes of wet processing.

Objectives of Scouring

- To make the fabric highly hydrophilic
- To remove impurities such as oils, waxes, gum, husks as nearly as possible
- To increase the absorbency of fabric or textile materials without physical and chemical damage
- To produce a clean material by adding alkali
- To prepare the fabric for subsequent process
- To remove non-cellulosic substance in case of cotton

Scouring can be carried out by two methods:

Saponification

The vegetable oil like glycerides of fatty acids, are present in the raw fabric. These oils are immiscible with water and hence they are heated with a solution of sodium hydroxide in water making the oil split up into its constituents-fatty acid and glycerine. Glycerine is mixable with water easily and the fatty acids react with sodium hydroxide present in the solution forming its sodium salt i.e. soap which is also soluble in water. Thus oil is removed from the fabric.

Emulsification

The wax and non saponifiable oils present in the fabrics are removed by emulsification. As the waxes and oils are not mixable in water a normal washing soap is used as an emulsifying agent. The soap makes an emulsion of the waxes and non saponifiable oils in water and brings them out.

The changes that occur during scouring are as follows :

- Saponifiable oils and free fatty acids are converted into soaps
- Pectins and pectoses are converted into soluble salts of pectic acid
- Proteins are degraded to simple soluble amino acids or ammonia
- Mineral matters are mostly dissolved
- Non- saponifiable oils are emulsified by the soluble soaps generated from the saponifiable oils
- Additive stains are removed
- Residual sizing materials are broken down into soluble products

The chemicals used in scouring are:

1. Caustic soda - to neutralize acidic materials, to saponify glycerides (waxes and oils), to solubilise silicates.
2. Surfactants - to reduce surface tension and minimize interfacial tension.
3. Detergents - to emulsify oil, fats, waxes and remove oil -borne stains.
4. Chelating agent - to deactivate metal ions.
5. Sodium silicate - to penetrate and break down lignins
6. Soda ash - to maintain pH
7. Solvent - to assist emulsification by dissolving oily materials.

1.3.2 Bleaching

Bleach is a chemical that removes colour or whitens a fabric via oxidation or reduction method. Many types of bleach have strong bactericidal properties, and are used for disinfecting and sterilizing.

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